

Bayesian Network Modeling of

NFL Game Stats & Market Signals

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3,990

NFL Games
15 Seasons

Introduction



PROBLEM STATEMENT

Two information sources exist for every NFL game — **historical game stats** and **market signals**.

**Do market signals encode predictive information
beyond what game stats already capture?**



WHY IT MATTERS

First formal test of conditional dependence between game fundamentals, market, and outcome.



SIGNIFICANCE

Either result is meaningful — **inefficiency found** or **closing line sufficiency confirmed**.



HISTORICAL FRAMING

2007–2022 data window, spanning the 2018 PASPA repeal — not a prescriptive system.

Literature Review

Group 1 BNs for Sports Prediction

Constantinou, Fenton & Neil (2012–13)

BN for soccer forecasting — game variables only, no market movement

Group 2 Stats + Odds Combined

Egidi, Pauli & Torelli (2018)

Hierarchical Bayes blending odds — no conditional dependence structure

Group 3 Market Efficiency / Line Movement

Simon (Mgmt. Sci.) (2024)

Line movement efficiency — no game stats integration

Moskowitz (J. Finance) (2021)

Sports betting as asset pricing — no conditional independence test

Szalkowski & Nelson (2012)

NFL opening/closing lines as predictors — never built as a model feature

GAP CONFIRMED

Walsh & Joshi (2024)

Systematic review of 219 papers — confirms no study combines all three silos

Objective

Research Question:

Does line movement contain predictive information beyond what game stats already capture?

Formally: *Is line movement conditionally independent of ATS outcome given game fundamentals?*

Line movement adds info

Markets contain signal beyond fundamentals — challenges efficient-market view.

Markets are efficient

Closing line absorbs all available info — confirmation is itself a publishable finding.

Either result advances the literature on market efficiency in NFL betting.

Methodology – Problem Formulation

Bayesian Network (BN)

DAG encodes domain assumptions:

historical stats → opening line → market reaction → outcome

Baseline: Naive Bayes

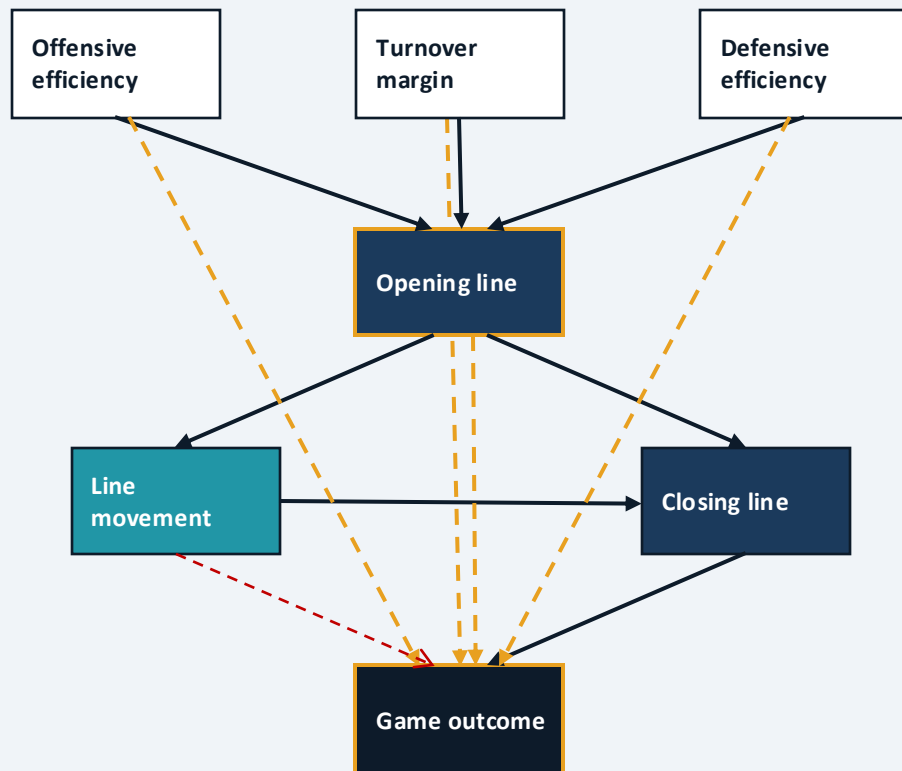
Assumes all features independent. If BN can't beat NB, conditional dependencies add no value → closing lines are efficient

Structure Learning

Hill Climbing to learn dependency structure from data

Three Test Modes

Stats-only · Market-only · Combined — to isolate independent signal from each source



Methodology - Evaluation Techniques

Classification Task — 3-class: **Cover** / **Loss** / **Push**

Gaussian Naive Bayes variant for continuous features

Accuracy

Baseline predictive
performance

Log Loss

Probabilistic calibration
quality

Brier Score

Mean squared error of
probabilities

AUC-ROC

One-vs-rest for binary
comparison

Ablation: Stats-only · Market-only · Combined

Isolates whether each source adds independent signal

Dataset

3,990

NFL Games

15

Seasons
2007-2022

6

Market Data Points / Game

2

Data Sources

Game Stats Sample (nflverse)

Season	Wk	Home	Away	H Score	A Score	EPA/play
2007	1	IND	NO	41	10	0.055
2007	1	BUF	DEN	14	15	-0.074
2007	1	CLE	PIT	7	34	-0.331
2007	1	GB	PHI	16	13	-0.082

Market Data Sample (Sportsbook Review)

Season	Home	Open Spread	Close Spread	ML Home
2007-08	Indianapolis	5.5	5.5	-240
2007-08	Houston	38.5	3.0	-200
2007-08	Buffalo	3.0	3.0	150
2007-08	Cleveland	4.5	5.0	195

Datasets joined by season + matchup (home & away teams) · 6 market data points per game including opening and closing spreads

Expected Challenges

Four anticipated risks and our mitigation strategy for each

1 Leakage enforcement

Rolling window must use only games strictly prior to game G; week 1 has no prior data.

2 BN structure sensitivity

Hill Climbing can stick in local optima; mitigated by initializing from a domain-informed DAG and running multiple restarts.

3 Class imbalance

Push (ATS tie) is a rare event; standard accuracy can mislead — use balanced metrics.

4 Continuous features

Standard BNs require discretization; Gaussian NB handles this natively, full BN needs care.

Questions & Discussion

Appendix

Timeline & Milestones

COMMITTED TIMELINE



Apr 26

✓ Done

Data pipeline complete (acquisition + preprocessing)



May 3

✓ Done

Project Proposal submitted



May 7

← Now

Proposal Presentation



May 17

MIN

Naive Bayes baseline implemented & evaluated



May 21

MIN

BN with Hill Climbing implemented



May 24

MIN

Ablation runs (stats-only / market-only / combined)



May 28

Paper draft complete



May 31

Final revisions & artifact submitted



Jun 4

Final Presentation

STRETCH GOALS — if time allows



May 26

DAG validation via PC algorithm

Work Division



Leland Weeks



Johnny Belichev



Ishant Somal

Key Takeaways

①

Novel Information Framework

First BN study to formally test conditional independence between NFL game statistics and betting market line movement

②

Rigorous Methodology

Hill Climbing structure learning + Naive Bayes baseline provides robust inference about market efficiency

③

Market Efficiency Test

If BN fails to outperform Naive Bayes, closing lines are formally efficient — a meaningful finding for sports analytics research

④

Rich, Dense Dataset

3,990 games over 15 seasons with EPA/play (down, distance, field position aware) and full market data from opening to closing